scenario: : You are developing a banking application that categorizes transactions based on the amount entered.

Logic:

1.Read the input numbers
2. If statement used greator then “1”,Print(“Doposite”)
3.elif using lessthen “1”,print(“Withdral”)
4.else,zero print(“No transaction”)

Scenario: A digital locker requires users to enter a numerical passcode. As part of a security feature, the system checks the sum of the digits of the passcode.

Logic:

1.Read the input number.
2.Convert the number into individual digits.
3. Initialize a sum variable to 0.
4. For each digit in the number, add it to the sum variable.
Print the sum of the digit

Scenario: A mobile payment app uses a simple checksum validation where reversing a transaction ID helps detect fraud.
Write logic to take a number and return its reverse.

Logic:

1.Read the input number
2.Numbers convert to string.
3.Revers the string.
4. Print, the reversed number.

Scenario: In a secure login system, certain features are enabled only for users with prime-numbered user IDs.
Write logic to check if a given number is prime

Logic:

1. **Read the input number.**
2. **If the number is less than 2, print "Not Prime".**
3. **Loop from 2 to the square root of the number:**
    - **If the number is divisible by any of these values, print "Not Prime" and exit.**
3. **If no divisors are found, print "Prime"**

Scenario: A scientist is working on permutations and needs to calculate the factorial of numbers frequently.
Write logic to find the factorial of a given number using recursion

Logic:

1. **Read the input number.**
2. **If the number is 0 or 1, return 1.**
3. **Else, return the number multiplied by the factorial of (number - 1).**
4. **Print the result.**

Scenario: A unique lottery system assigns ticket numbers where only Armstrong numbers win the jackpot.

Write logic to check whether a given number is an Armstrong number.

Logic:

1. **Read the input number.**
2. **Count the number of digits.**
3. **Initialize a sum variable to 0.**
4. **For each digit in the number:**
   - **Raise the digit to the power of the total number of digits.**
   - **Add the result to the sum variable.**
5. **If the sum is equal to the original number, print "Armstrong Number"**

Scenario: A password manager needs to strengthen weak passwords by swapping the first and last characters of user-generated passwords.

Logic:

1. **Read the input string.**
2. **If the string length is less than 2, print the string as is.**
3. **Swap the first and last characters while keeping the middle part unchanged.**

Print the modified string

scenario:A low-level networking application requires decimal numbers to be converted into binary format before transmission.

Write logic to convert a given decimal number into its binary equivalent.

Logic:

1. **Read the input decimal number.**
2. **Initialize an empty string for binary representation.**
3. **While the number is greater than 0:**

   .Divide the number by 2 and store the reminder

   . Add the remainder to the binary string.

   . Update the number by dividing it by 2.
4. **Reverse the binary string.**
5. **Print the binary representation.**

Scenario: A text-processing tool helps summarize articles by identifying the most significant words.

Write logic to find the longest word in a sentence.

Logic:

1. Read the input sentence.
2. Split the sentence into individual words.
3. Initialize a variable to store the longest word.
4. Loop through each word:
5. if the current word is longer than the stored longest word, update the longest word.
6. Print the longest word.

Scenario: A plagiarism detection tool compares words from different documents and checks if they are anagrams (same characters but different order).

Write logic to check whether two given strings are anagrams.

Logic:1. Read the two input strings.

2. Remove spaces and convert both strings to lowercase.

3. Sort the characters of both strings.

If the sorted versions of both strings are identical, print "Anagram".

4. Else, print "Not an Anagram".
